

Gravitational Waves and Numerical Relativity: Example Sheet 2

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1. Series expansion

Consider the leading-order expansion of the functions

$$\begin{aligned}\gamma(u, r, \theta) &= c(u, \theta)r^{-1} + \mathcal{O}(r^{-2}), \\ \beta(u, r, \theta) &= H(u, \theta) - c(u, \theta)r^{-2} + \mathcal{O}(r^{-3}),\end{aligned}$$

and the second main equation of the Bondi formalism,

$$\partial_r \left[r^4 e^{2(\gamma-\beta)} \partial_r U \right] - 2r^2 \left[\partial_r \partial_\theta \beta - \partial_r \partial_\theta \gamma + 2\partial_r \gamma \partial_\theta \gamma - 2 \cot \theta \partial_r \gamma - 2 \frac{\partial_\theta \beta}{r} \right] = 0, \quad (1)$$

Taking into account functions of integration, show that the leading-order behaviour of the function $U(u, r, \theta)$ is given by

$$U(u, r, \theta) = L(u, \theta) + 2e^{2H(u, \theta)} \partial_\theta H(u, \theta) r^{-1} + \mathcal{O}(r^{-2}). \quad (2)$$

2. The metric and its determinant

Let $g_{\alpha\beta}$ be a metric of Lorentzian signature $g_{\alpha\beta}$ and $g \equiv \det g_{\alpha\beta}$ its determinant. Show that

$$\begin{aligned}\frac{\partial g}{\partial g_{\alpha\beta}} &= g g^{\alpha\beta}, \\ \frac{\partial g}{\partial g^{\alpha\beta}} &= -g g_{\alpha\beta},\end{aligned}$$

where $g^{\alpha\beta}$ denotes the inverse metric. Conclude that the derivative of the determinant g can be written as

$$\partial_\alpha g = g g^{\mu\nu} \partial_\alpha g_{\mu\nu} = -g g_{\mu\nu} \partial_\alpha g^{\mu\nu} = 2g \Gamma_{\mu\alpha}^\mu.$$

3. Absence of conical singularities

The Bondi metric for an axisymmetric spacetime that is invariant under azimuthal reflection is

$$ds^2 = \left(-\frac{V}{r} e^{2\beta} + U^2 r^2 e^{2\gamma} \right) du^2 - 2e^{2\beta} du dr - 2Ur^2 e^{2\gamma} du d\theta + r^2 (e^{2\gamma} d\theta^2 + e^{-2\gamma} \sin^2 \theta d\phi^2),$$

where β , γ , U and V are functions of the coordinates (u, r, θ) . Consider the case of asymptotically flat spacetimes with no incoming radiation, so that the series expansion of γ at large radius r is given by

$$\gamma(u, r, \theta) = c(u, \theta)r^{-1} + \mathcal{O}(r^{-3}),$$

where $c(u, \theta)$ is the Bondi news function.

Show that for spacetimes with no conical singularity on the polar axis, the news function satisfies

$$\lim_{\theta \rightarrow 0} c = \lim_{\theta \rightarrow \pi} c = 0.$$

[Hint: Consider spheres of constant u and r in the limit of large r . Compute the proper circumference and the proper radius of circles with small constant $\theta = \Delta\theta$ around the North Pole and show that their ratio equals 2π if and only if the first of the above conditions is satisfied. Repeat the same for the South Pole.]

4. The Bondi mass

Let $(\mathcal{M}, g)$ be an asymptotically flat, axisymmetric spacetime that is invariant under azimuthal reflection and contains no conical singularities. In the Bondi formalism for such spacetimes, the mass aspect $M(u, \theta)$ arises as one of the functions of integration. By the supplementary equations, the evolution of the mass aspect is given in terms of the Bondi news $c(u, \theta)$ by

$$\partial_u M = -\partial_u c + \frac{1}{2} \partial_\theta^2 \partial_u c + \frac{3}{2} \cot \theta \partial_\theta \partial_u c - (\partial_u c)^2.$$

The Bondi mass is defined as the integral of the mass aspect,

$$m(u) := \frac{1}{4\pi} \int_0^\pi \int_0^{2\pi} M \sin \theta d\phi d\theta.$$

Using the result from question 3, show that the Bondi mass evolves according to

$$\partial_u m = -\frac{1}{2} \int_0^\pi (\partial_u c)^2 \sin \theta d\theta.$$

Briefly interpret this result.

5. Spatial covariant derivative

Let $(\mathcal{M}, g)$ be a globally hyperbolic spacetime with Levi Civita connection ∇_α and spatial hypersurface Σ (i.e. the gradient dt is timelike), and let $T^{\alpha\cdots}_{\beta\cdots}$ be a rank $\binom{r}{s}$ tensor that is tangent to Σ in all components, i.e. $T^{\alpha\cdots}_{\beta\cdots} n_\alpha = \cdots = T^{\alpha\cdots}_{\beta\cdots} n^\beta = \cdots = 0$. The three-dimensional or spatial covariant derivative of T is defined as the rank $\binom{r}{s+1}$ tensor

$$D_\mu T^{\alpha\cdots}_{\beta\cdots} := \perp^\rho_\mu \perp^\alpha_\sigma \perp^\tau_\beta \cdots \nabla_\rho T^{\sigma\cdots}_{\tau\cdots}.$$

(i) Show that D_μ satisfies the defining criteria for a covariant derivative, i.e. for spatial vector fields X, Y, V and a scalar function f on Σ the following conditions hold,

- (1) $D_\mu f = \partial_\mu f,$
- (2) $D_{fX+gY} V = f D_X V + g D_Y V,$
- (3) $D_X V + D_X W = D_X (V + W),$
- (4) $D_X (fV) = f D_X V + (D_X f) V.$

(ii) Show that D_μ is compatible with the three-dimensional (spatial) metric $\gamma_{\alpha\beta}$, i.e. $D_\mu \gamma_{\alpha\beta} = 0$.

(iii) The torsion tensor associated with D_μ is defined by

$$T : (X, Y) \mapsto T(X, Y) = D_X Y - D_Y X - [X, Y]$$

$$\text{or } T_{\mu\nu}{}^\alpha X^\mu Y^\nu = X^\mu D_\mu Y^\alpha - Y^\mu D_\mu X^\alpha - [\mathbf{X}, \mathbf{Y}]^\alpha,$$

where $\mathbf{X}, \mathbf{Y}$ are spatial vector fields and $[\mathbf{X}, \mathbf{Y}]^\alpha = X^\mu \partial_\mu Y^\alpha - Y^\mu \partial_\mu X^\alpha$ is the commutator of $\mathbf{X}$ and $\mathbf{Y}$. Show that D_μ is torsion free, i.e. its torsion tensor vanishes.

6. Lie derivative of the projector

Given a hypersurface Σ of a globally hyperbolic spacetime with unit normal n^α and projector $\perp^\alpha_\beta = \delta^\alpha_\beta + n^\alpha n_\beta$, show that

$$\mathcal{L}_n \perp^{\alpha\beta} = n^\alpha a^\beta + n^\beta a^\alpha + 2K^{\alpha\beta},$$

$$\mathcal{L}_n \perp^\alpha_\beta = n^\alpha a_\beta,$$

$$\mathcal{L}_n \perp_{\alpha\beta} = -2K_{\alpha\beta},$$

where $\mathcal{L}_n$ is the Lie derivative along n^α . Conclude that for a spatial tensor $T_{\alpha\beta}$,

$$\mathcal{L}_n T_{\alpha\beta} = \perp^\mu_\alpha \perp^\nu_\beta \mathcal{L}_n T_{\mu\nu},$$

i.e. the tensor's Lie derivative along n is also spatial.

7. Ricci equation

Let Σ be a spatial hypersurface of a globally hyperbolic spacetime with unit normal n^α , acceleration vector $a_\alpha = n^\mu \nabla_\mu n_\alpha$ and extrinsic curvature $K_{\alpha\beta}$.

(i) By projecting the (spacetime) Ricci identity applied to the unit normal,

$$\nabla_\rho \nabla_\sigma n^\mu - \nabla_\sigma \nabla_\rho n^\mu = R^\mu{}_{\nu\rho\sigma} n^\nu,$$

twice onto space and once onto time, show that

$$\perp_{\alpha\mu} n^\nu \perp^\rho_\gamma n^\sigma R^\mu{}_{\nu\rho\sigma} = -K_{\alpha\sigma} K^\sigma{}_\gamma + D_\gamma a_\alpha + a_\alpha a_\gamma + \perp^\mu_\alpha \perp^\rho_\gamma n^\sigma \nabla_\sigma K_{\rho\mu}. \quad (\dagger)$$

(ii) Compute the spatial projection of the Lie derivative of the extrinsic curvature along the unit normal, $\perp^\mu_\alpha \perp^\nu_\beta \mathcal{L}_n K_{\mu\nu}$, to substitute for the last term on the right-hand side of Eq. ($\dagger$), and thus derive the Ricci equation

$$\perp^\mu_\alpha n^\nu \perp^\rho_\gamma n^\sigma R_{\mu\nu\rho\sigma} = \mathcal{L}_n K_{\alpha\gamma} + \frac{1}{\alpha} D_\alpha D_\gamma \alpha + K_{\rho\gamma} K_\alpha{}^\rho.$$

You may use without proof that $a_\mu = D_\mu(\ln \alpha)$, where α is the lapse function.

8. The evolution equation for the energy density

The energy- and momentum density and stress associated with the energy momentum tensor $T_{\alpha\beta}$ are defined as

$$\rho := T_{\mu\nu} n^\mu n^\nu, \quad j_\alpha := -\perp^\mu_\alpha T_{\mu\nu} n^\nu, \quad S_{\alpha\beta} := \perp^\mu_\alpha \perp^\nu_\beta T_{\mu\nu},$$

$$\Leftrightarrow T_{\alpha\beta} = \rho n_\alpha n_\beta + j_\alpha n_\beta + n_\alpha j_\beta + S_{\alpha\beta},$$

where n^μ denotes the unit normal of a spatial hypersurface Σ and $\perp^\mu_\alpha = \delta^\mu_\alpha + n^\mu n_\alpha$. The equation for conservation of energy-momentum is $\nabla_\mu T^\mu{}_\alpha = 0$.

By projecting the energy conservation law onto n^α , derive the evolution equation for the energy density,

$$\mathcal{L}_n \rho = -2j^\mu D_\mu (\ln \alpha) + \rho K + S^{\mu\nu} K_{\mu\nu} - D_\mu j^\mu.$$

Show that in coordinates adapted to the spacetime foliation, this equation becomes

$$\partial_t \rho = \beta^m \partial_m \rho - 2j^m D_m \alpha + \alpha (\rho K + S^{mn} K_{mn} - D_m j^m).$$

Here K is the trace of the extrinsic curvature, α the lapse and β^μ the shift vector.

9. Strong hyperbolicity

Consider the partial differential equation

$$\mathbf{A} \partial_t \mathbf{u} + \mathbf{P}^i \partial_i \mathbf{u} + \mathbf{C} \mathbf{u} = 0 \quad (\text{sum over } i), \quad (\star)$$

for a function $\mathbf{u} : \Omega \subset \mathbb{R}^{d+1} \rightarrow \mathbb{R}^N$, where $\mathbf{A}, \mathbf{P}^i, \mathbf{C}$ are real $N \times N$ matrices and $\mathbf{A}$ is invertible. This PDE is weakly hyperbolic if for all vectors $\hat{k}^i$ with $|\hat{k}| = 1$, all Eigenvalues of

$$\mathbf{Q}(\hat{k}^i) := -\mathbf{A}^{-1} \mathbf{P}^m \hat{k}_m$$

are real. The PDE is strongly hyperbolic if furthermore $\mathbf{Q}(\hat{k}^i)$ is diagonalizable for all $\hat{k}^i$.

(i) Let $\mathbf{J}_2 = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$. Show that

$$e^{i\mathbf{J}_2 t} = e^{i\lambda t} \begin{pmatrix} 1 & it \\ 0 & 1 \end{pmatrix}.$$

(ii) Assume that the PDE $(\star)$ is weakly hyperbolic but that the Jordan normal form of $\mathbf{Q}(\hat{k}^i)$ contains a Jordan block of the form $\mathbf{J}_2$. Argue why for such a PDE, there exists no regular function $f(t)$ such that

$$\left\| e^{i\mathbf{M}t} \right\| \leq f(t), \quad (\ddagger)$$

where $\mathbf{M} = \mathbf{A}^{-1}(-\mathbf{P}^m k_m + i\mathbf{C})$.